

Outcome	First heads is on toss #1	First heads is on toss #2	First heads is on toss #3	First heads is on toss #4	First heads is on toss #5	
Payoff	\$2	\$4	\$8	\$16	\$32	
Probability	1/2	1/4	1/8	1/16	1/32	

$$1+1+1+1+1+\quad 1+1+1+1+1+1+1+1+1+1+1+1+1+1+1+1+\dots$$

But it follows from this result, plus the rule of expected utility, that **you would be rational to pay any finite amount of money to have the chance to play this game once**. But this seems clearly mistaken. Does this show that expected utility considerations fall apart in contexts with infinite rewards?

Suppose that we set an upper bound of 100 coin flips on the game, so that if you get to the 100th flip you get $\$2^{100}$ (a very large number) no matter how the coin comes up. Then the expected utility of playing will be $\$100$. Would you pay $\$99$ to play this game?

Most would say not. One possibility is that this is explained by a combination of **risk aversion** and **decreasing marginal utility**. Could these also play a role in the evaluation of Pascal's wager?

The expected utility of playing = the sum of probability $\times$ payoff for each of the infinitely many

possible outcomes. So, the expected utility of playing equals the sum of the infinite series

[illegible]

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